EDITORIAL

SCIENCE BEHIND THE STUDY

Personalized Gene Editing to Treat an Inborn Error of Metabolism

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Musunuru et al.[1] report in the *Journal* the development of a personalized gene-editing strategy to treat a single infant with carbamoyl-phosphate synthetase 1 (CPS1) deficiency: they treated the infant when he was approximately 7 months of age. As discussed by Marks in an accompanying editorial,[2] this article is a milestone in the evolution of personalized therapies for rare and ultrarare inborn errors of metabolism. However, one must be sensitive to concerns about the evidentiary standards applied in such early interventions.

WHAT IS CPS1?

CPS1 is a mitochondrial enzyme that catalyzes the first and rate-limiting step of the urea cycle: the conversion of ammonia and bicarbonate to carbamoyl phosphate (Fig. 1). This is a crucial detoxification step that occurs primarily in hepatocytes. Without functional CPS1, ammonia accumulates rapidly in the bloodstream and causes hyperammonemia, which is toxic to the brain and can lead to coma or death if untreated.

CPS1 deficiency is an autosomal recessive inborn error of metabolism and one of the most severe and rarest urea-cycle disorders: its estimated incidence is approximately 1 in 800,000 to 1 in 1,300,000 live births. It typically manifests as profound hyperammonemia in the first 24 to 48 hours after birth. Clinical interventions are limited and may include dialysis, ammonia scavengers, protein restriction, and, later in life, liver transplantation, but neurologic outcomes are often poor. Gene editing presents a compelling therapeutic opportunity for infants with CPS1 deficiency and certain other urea-cycle disorders.

KEY CONCEPTS

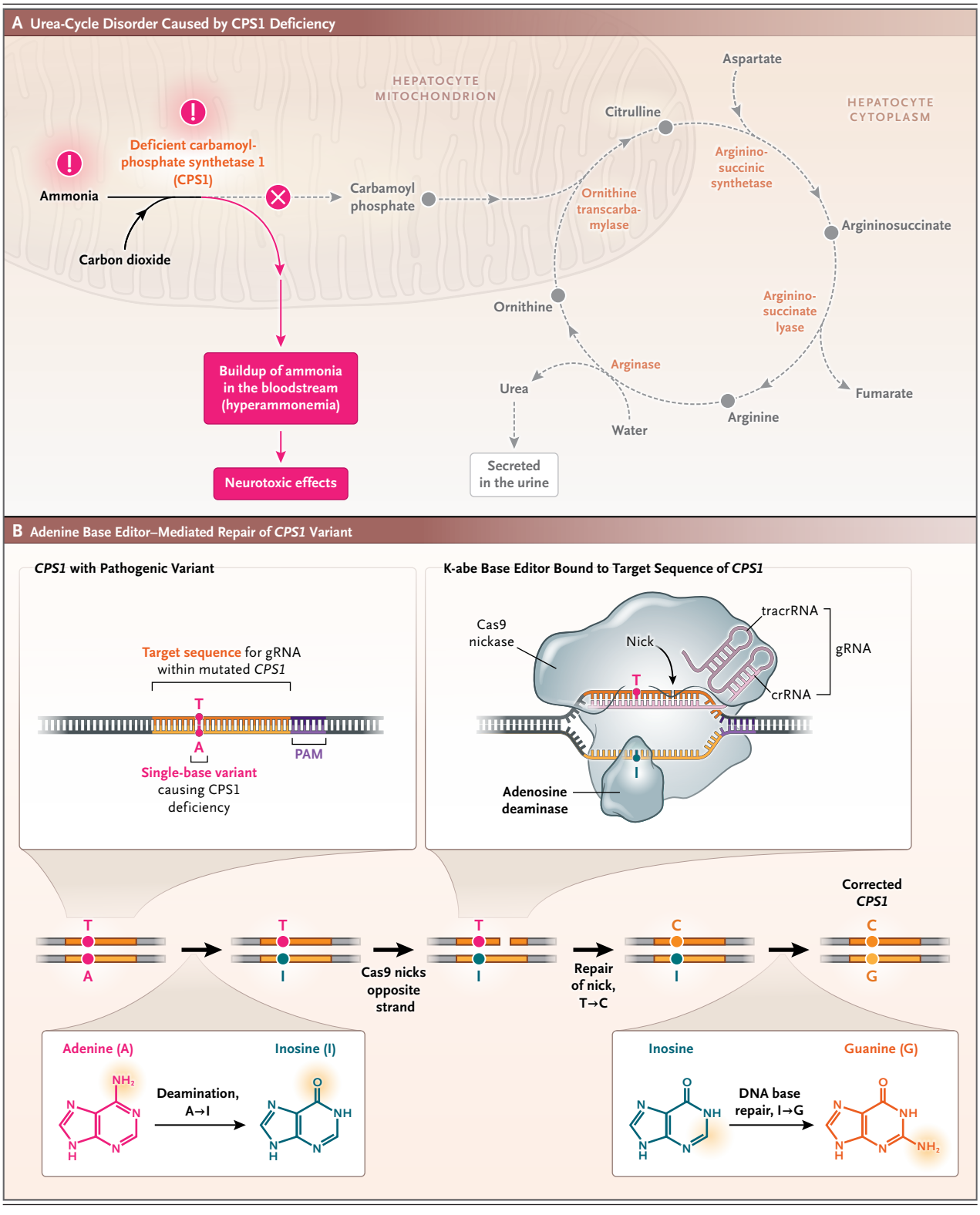
Base-pair
Two nitrogenous bases paired together in double-stranded DNA by weak bonds. Specific pairing of these bases (adenine with thymine and guanine with cytosine) facilitates accurate DNA replication.

CRISPR-Cas9 editing
A process in which clustered regularly interspaced short palindromic repeats (CRISPR) — small remnants of viral DNA from previous viral infection that are embedded in bacterial DNA — generate guide RNAs, which are used together with a DNA-cutting enzyme (Cas9) by the bacterium to defend against viral infection. With synthetic guide strands and an exogenous Cas enzyme, this technology can be used in eukaryotic cells to modify DNA or RNA for the purpose of gene editing.

Hepatocyte
A primary cell of the liver. Hepatocytes make up 70 to 80% of the liver's mass and are involved in many of the liver's key functions, including detoxification and the metabolism of carbohydrates and lipids. In addition, they are involved in the production of proteins, such as albumin, clotting factors and various complement factors, and bile. They can be infected by hepatotropic viruses including hepatitis A, B, C, D, and E viruses. The liver has a remarkable ability to regenerate because hepatocytes can reenter the cell cycle from a quiescent phase. Nevertheless, in the context of chronic liver disease, such as chronic hepatitis B virus infection, they are slowly displaced by fibrous tissue, eventually leading to cirrhosis, liver failure, and liver cancer.

Nonsense variant
A single DNA nucleotide that has been altered such that an amino acid codon becomes a stop codon, leading to truncation of the encoded protein.

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Figure 1 (facing page). A Strategy to Treat CPS1 Deficiency through Base Editing.

Deficiency of carbamoyl-phosphate synthetase 1 (CPS1) is a recessive urea-cycle disorder caused by damaging or null variants in the gene *CPS1*. The CPS1 enzyme catalyzes the first step in the urea cycle, which takes place in the hepatocyte (Panel A). This step involves the combination of ammonia and carbon dioxide to produce carbamoyl phosphate, thus converting a neurotoxin (ammonia) into a substrate of the urea cycle (carbamoyl phosphate). The product of this cycle, urea, is excreted from the body in the urine. In the absence of adequate levels of CPS1, ammonia levels increase. An infant described by Musunuru et al.[1] presented with symptoms within 2 days after birth. Within 7 months, Musunuru et al. generated and carried out preclinical studies of an adenine base editor (kayjayguran abengcemeran, or k-abe) (Panel B), designed to correct one of the infant's pathogenic *CPS1* variants. The editor is made up of a modified clustered regularly interspaced short palindromic repeats–associated protein 9 (Cas9) enzyme (which does the editing) and a guide RNA (gRNA, which guides the Cas9 enzyme to the mutant *CPS1* nucleotide). In preclinical experiments, k-abe changed (or "edited") the mutant adenine into an inosine (which resembles guanine) on the noncoding strand, which resulted in the correction of the thymine to cytosine on the coding strand and permitted synthesis of full-length CPS1. Reduced levels of ammonia in the infant's plasma after treatment with k-abe is consistent with correction of the pathogenic *CPS1* variant. The term crRNA denotes CRISPR (clustered regularly interspaced short palindromic repeats) RNA, PAM protospacer adjacent motif, and tracrRNA trans-activating CRISPR RNA.

HOW TO TREAT CPS1 DEFICIENCY WITH A GENETIC MEDICINE?

CPS1 is encoded by a particularly large gene, which complicates its delivery with common vectors such as adeno-associated virus. In addition, the risk of vector dilution due to hepatocyte turnover in neonates may further reduce the efficacy of conventional gene therapy.

Moreover, because only a handful of patients worldwide have the same ultrarare disease, it is challenging to assemble robust preclinical and clinical data. Natural-history registries are therefore valuable in this context. Among patients with the same variant, there may be variable expressivity, and clinical presentation may vary widely, which complicates outcome measures. And CPS1 deficiency is genetically heterogeneous: it is caused by a variety of different mutations. With respect to gene editing, each mutation would require its own unique editing strategy, the development of which is costly and can take years to identify and sufficiently customize.

WHAT IS BASE EDITING?

Base editing is a method that can be used to install single variants into the genome of live cells in a targeted, precise, and efficient manner. Thus, one can revert a pathogenic stop codon to the "original," nonmutant sequence, allowing the synthesis of a full-length, functional protein. Musunuru et al. used an adenine base editor (ABE), which comprises an enzyme (a clustered regularly interspaced short palindromic repeats–associated protein 9 [Cas9] nickase [Cas9n]–adenosine deaminase fusion) and a piece of RNA called a guide RNA (gRNA). The Cas9n enzyme is guided to a specific 20 base-pair (see Key Concepts) protospacer sequence (in this case, the region of the patient's paternal *CPS1* gene that harbors the mutation) with the gRNA. The protospacer must be next to a three-base motif, called the protospacer adjacent motif (PAM), which the Cas9n protein recognizes and binds tightly to. Once bound, the adenosine deaminase domain deaminates adenines within the protospacer, leading to the conversion of a specific A–T base pair into a G–C base pair. Base editing is therefore more precise than clustered regularly interspaced short palindromic repeats [CRISPR]–Cas9 editing, which disrupts genes by cutting the DNA rather than targeting a single base pair for mutation.

HOW CAN IT TREAT CPS1 DEFICIENCY?

Musunuru et al. developed a base-editing strategy to correct the *CPS1* Q335X "stop" variant, a nonsense G-to-A variant on the noncoding strand. They started out by testing several different gRNAs that target overlapping protospacers. With each gRNA, the pathogenic A variant is in a slightly different position and orientation with respect to the positioning of the adenosine deaminase domain, once the ABE has bound to the PAM and protospacer.

Increasing the potential for patient-specific customization, each gRNA can be combined with an ABE variant: ABE variants differ in their preferences for specific DNA motifs. Numerous

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ABE:gRNA combinations were individually generated and tested to identify the combination with the greatest efficiency and precision. Further complicating the testing process is that it must be carried out with the use of cells that harbor the pathogenic variant and that are the same or a similar cell type to the target tissue (in the case of CPS1 deficiency, a hepatocyte) and are easy to culture and grow in the laboratory. For some rare diseases, it can take months to a year to generate such a cell line.

HOW WAS GENE-EDITOR CUSTOMIZATION ACCELERATED?

To obtain a suitable cell line for testing ABE:gRNA combinations, Musunuru et al. inserted a 100 base-pair segment representing the patient's mutated *CPS1* (generated through DNA-synthesis technology) into the genome of a human hepatocyte cell line. Overall, the generation of the cell line and ABE:gRNA screening took 2 months to complete. Preclinical studies of the adenine base editor (named kayjayguran abengcemeran, or k-abe) were then conducted in two different transgenic mouse models and in nonhuman primates.

WHAT'S NEXT?

The ability to intervene at the genomic level and to directly correct the underlying genetic defect in a patient's liver shows the promise of CRISPR-based medicine. Limitations of the study include the fact that the mouse model, although carrying the patient's pathogenic *CPS1* variants, does not have a CPS1 deficiency phenotype and so is unsuitable for testing therapeutic efficacy.

This study must be balanced against the current limitations of an N-of-1 experience in a rare or ultrarare condition. Although the authors noted clinical stabilization in the short term, the absence of direct molecular confirmation of gene editing by means of liver biopsy leaves questions unanswered, such as the durability of the therapeutic effect, the extent of mosaicism of editing, and the risks of off-target events or immune responses. Longer-term follow-up of this patient will be critical to obtaining answers.

The development of minimally invasive biomarkers in lieu of liver biopsy to assess editing success and expansion to small, well-controlled cohorts should be considered. One could consider ureagenesis studies that use stable isotope tracers, cell-free DNA assays, or quantitative metabolomics. Further customization of delivery platforms and enhancement of editing efficiency will be key to translating this singular success into reproducible and scalable therapy. Moreover, the path from proof of concept to standard of care will require careful ethical and regulatory stewardship. For patients with CPS1 deficiency and similar ultrarare disorders, the findings of Musunuru et al. offer hope and yet require validation through treatment of the second patient, the third, and beyond.

Disclosure forms provided by the authors are available with the full text of this editorial at NEJM.org.

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1. Musunuru K, Grandinette SA, Wang X, et al. Patient-specific in vivo gene editing to treat a rare genetic disease. N Engl J Med. DOI: 10.1056/NEJMoa2504747.
2. Marks P. Progress in the development of N-of-1 therapy. N Engl J Med. DOI: 10.1056/NEJMe2505704.

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